

5	(a) work done bringing/moving per unit positive charge from infinity (to the point)	M1 A1	[2]
	(b) (i) slope/gradient (of the line/graph/tangent) (allow dV/dx , but not $\Delta V/\Delta x$ or V/x) (allow potential gradient) (negative sign not required)	B1	[1]
	(ii) maximum at surface of sphere A or at $x = 0$ (cm) zero at $x = 6$ (cm) then increases but in opposite direction (any mention of attraction max. 2/3)	B1 B1 B1	[3]
	(c) (i) M shown between $x = 5.5$ cm and $x = 6.5$ cm	B1	[1]
	(ii) 1. $\Delta V = (570 - 230) = 340$ V (allow 330 V to 340 V)	A1	[1]
	2. $q(\Delta)V = \frac{1}{2}mv^2$ or change/loss in PE = change/gain in KE or $\Delta E_K = \Delta E_P$	B1	
	$4.8 \times 10^7 \times 340 = \frac{1}{2}mv^2$ $v^2 = 3.26 \times 10^{10}$	C1	
	$v = 1.8 \times 10^5 \text{ m s}^{-1}$ (not 1 s.f.)	A1	[3]